Tobias Mwangi-Hartley

applicant.inbox@example.com | +44 7700 900123

PROFESSIONAL SUMMARY

Data scientist with experience developing predictive models, engineering features and building scheduled and batch-scoring workflows. Uses Python, SQL and GCP tools including BigQuery and Vertex AI Workbench. Seeking to grow into machine learning engineering and deepen production ML experience.

CORE SKILLS

Programming and analysis: Python, pandas, scikit-learn, SQL, R
Machine learning and statistics: LightGBM, XGBoost, SHAP, statsmodels, experimental design, Bayesian hierarchical models
Data and workflow tools: BigQuery, Vertex AI Workbench, dbt, Looker, R Markdown
Engineering practices: pytest unit testing, Dockerfile authoring, Streamlit
Additional learning: PyTorch, torchvision

EXPERIENCE

Data Scientist | Thistlebox | 2022-Present
Work across experimentation, forecasting and churn modelling. In 2025, began writing pytest unit tests for forecasting feature code, reaching approximately 60% coverage. Delivered internal training, "Stats for PMs," three times.

Junior Data Analyst | Firth Energy Retail | 2020-2022
Segmented customers and analysed tariffs using R and SQL. Automated monthly regulator reports with R Markdown, saving the team approximately two days each month.

SELECTED PROJECTS

Recipe Demand Forecasting | Thistlebox | 2022-2024

- Built weekly order forecasts six weeks ahead for each recipe and region using LightGBM with calendar, price and menu-position features.
- Ran forecasts as a scheduled Vertex AI Workbench notebook job, writing results to BigQuery for operations use in Looker.
- Improved WAPE from 23% to 14%; estimated food waste reduction was 9%.

Sequential A/B Testing Calculator | Thistlebox | 2023

- Built a Streamlit app for planning and analysing experiments with sequential boundaries, used by approximately 25 people in product and growth.
- Authored the app's Dockerfile; the data platform team handled hosting and deployment on Cloud Run.

Churn Propensity and Retention Targeting | Thistlebox | 2024-2025

- Developed an XGBoost churn model scored weekly in BigQuery; retention offers were sent to top-decile customers.
- In a 10% holdout validation, churn was down 2.1 percentage points in the treated group.

Image Classification Course Project | Personal | 2025

- Fine-tuned a small CNN in PyTorch on a public food images dataset, achieving approximately 80% accuracy as part of an online deep learning course.

EDUCATION

MSc in Statistics
Dissertation: Bayesian hierarchical models